Design and development of Automatic Bengali Number Plate Recognition (ABNPR) by Supervised Machine Learning



This report is submitted for the partial fulfillment of the requirement for the degree of B.Sc. (Hon's) in Computer Science and Engineering (CSE) under National University.



Submitted By:

Registration No : 17502004685

Session : 2017-18

Department of Computer Science and Engineering

Institute of Science and Technology

National University, Bangladesh

DECLARATION

I hereby declare that I have completed the work of this project under the supervision of **Tania Sultana,** Lecturer, Department of Computer Science and Engineering (CSE), Institute of Science and Technology (IST), affiliated with the National University of Bangladesh. This is also declared to me that neither this project nor any part of this project has been used or submitted elsewhere for any kind of degree or award.

Signature

…….

Registration No : 17502004685

Session : 2017-18

Department of CSE

Institute of Science and Technology

APPROVAL

This project titled “Design and Development of Automatic Bengali Number Plate Recognition (ABNPR) by Supervised Machine Learning”, submitted by **Ridoy Kanto Joy**, Registration No: 17502004685 to the Department of Computer Science and Engineering, Institute of Science and Technology (IST), Dhaka, Bangladesh has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering under National University and approval as to its style and contents.

Signature of Internal Examiner	**Signature of External Examiner**
……	1.
Tania Sultana	
(Supervisor)	2.
Lecturer	
Institute of Science and Technology	

ACKNOWLEDGMENT

This is my humble attempt to present gratitude in preparing this report. I have truly drawn upon my own experience as a student of computer science. This project would not have been possible without the dedication and contributions of many individuals.

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ABSTRACT

Automatic bengali number plate recognition (ABNPR) is a system that can detect and read the license plates of vehicles in Bengali script without human intervention. ABNPR has various applications such as traffic management, security, parking control, vehicle tracking and law enforcement. However, ABNPR faces many challenges due to the variations in number plate formats, fonts, colors, sizes and orientations, as well as the presence of noise, occlusion and distortion in the images. In this paper, we propose a novel segmentation-free neural network architecture for License Plate Recognition(LPR) based on Inception-ResNet v2 [22]. With the deep learning infrastructure provided by this, the network learning is deeper and requires fewer parameters; that is, it is efficient and fast, compared to other machine learning methods. 1080 images of Bangladeshi vehicle number plate datasets are used to verify the proposed method, showing that it achieves 99 % accuracy with an average fps (frames per second) of over 75, indicating that it is applicable under different environments and challenging conditions without any implementation change. Our system outperforms the existing methods in terms of accuracy and efficiency and can be used for ABNPR applications.

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Chapter 1

INTRODUCTION

1.1 Introduction

Automatic Bengali Number Plate Recognition System (ABNPR) is an essential stage for the automation of traffic system. Use of vehicles is getting increased in today's era that is why traffic control is being tough. It is hard to store and maintain the record of vehicles manually. Automatic Number Plate Recognition System can be used for better control of vehicles and for store and maintain the record of vehicles automatically.

1.2 Objectives

One of the first challenges we encountered while working in this research was the lack of a publicly accessible dataset for Bangla license plates. In addition, unlike the most popular one-line license plate format around the world, Bangla license plates have two lines. Another challenge that we faced is that car owners were reluctant to provide us permission to take pictures of their car license plates. That's why it took long time to prepare a database of 250 Bangla license plate images. There are normal, bright, dark, distant, blur, skewed, tilted, & noisy images in our database. We have also prepared Bangla templates (alphabetic, numeric and word) for the recognition purpose.

Pre-processing, processing, recognition and authentication of the license plate image are the four key components of the proposed system. The captured image is read, resized, and converted to gray-scale during pre-processing. Then, in the processing part, the plate location is extracted, followed by a segmentation process to separate the characters of the plate. Plate characters are recognized using correlation-based template matching. The license plate number saved in a text file is matched with a database prepared for the authentication purpose of the vehicles.

Number plate detection is identifying the part of the car that is predicted to be the number plate. Recognition is identifying the values that make up the license plate. License plate detection and recognition is the technology that uses computer vision to detect and recognize a license plate from an input image of a car. On roads, it is used to identify the cars that are breaking the traffic rules. In security, it is used to capture the license plates of the vehicles getting into and out of certain premises. In parking lots, it is used to capture the license plates of the cars being parked. The list of its applications goes on and on.

The Bangladesh Road Transport Authority (BRTA) is theonly authoritative body in Bangladesh to issue various typesof license plates for vehicles. The current version of the BRTAlicense plates started in 1973 but the digitization of this kindof license plates stared in 2012. Now it is a requirement forthe vehicles to have digital license plates on both the frontand rear sides of a vehicle, where the rear license plate ispermanently attached. Figure refers to the classes of licenseplates in Bangladesh.

Vehicle Category	Number Plate Color	Character color	Number Plates
Commercial	Green	Black	
Private	White	Black	

Figure 1.1: Different classes of number plate.

In the BRTA guideline, Bengali alphabet and Bengali nu-merals are used in the vehicle license plates. BRTA format of the digital license plates is divided into two lines. The firstline presents the “city – vehicle class letter” while the secondline shows the “class number – vehicle number”. Figure isan example of the BRTA standard license plate.

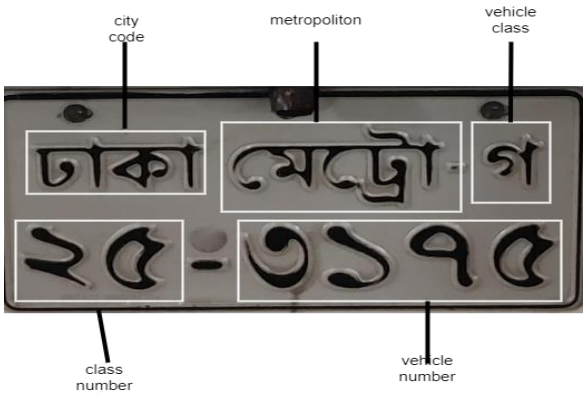


Figure 1.2: BRTA standard number plate.

1.3 Motivation

Automatic bengali number plate recognition can help improve the traffic management and security in Bangladesh, where the number of vehicles is increasing rapidly and manual process of checking the number plates is inefficient.

It also can enable various applications such as automated toll collection, parking lot management, stolen vehicle detection, vehicle tracking and law enforcement.

ABNPR can benefit from the recent advances in deep learning and computer vision, such as convolutional neural network (CNN), object detection models (such as YOLO, Inception ResNet V2) and image processing techniques (such as Otsu's thresholding). These methods can help improve the accuracy and efficiency of the system.



Figure 1.3: Applications of ABNPR.

1.4 Scope of the Project

Number plate recognition is realized by acquiring images of either the front or the rear of vehicles with cameras and then by image processing to identify license plates. It consist of three main stages. First one is Number Plate Identification & Localization in this segment the visual of the scene is improved with is image processing. Second is Character Segmentation in which characters segmented from the detected number plate for retaining the useful information to the system so that further processing can take place. Third is OCR Optical Character Recognition in which text is transferred into encoded text information. A feed forward Artificial Neural Network is used which is based on OCR Algorithms.

1.5 Organization of the Project

The 1st chapter "Introduction" contains the information about project motivations, objectives, scope of the project.

The 2nd chapter "Related literature" contains the information about "Automatic Bengali number plate recognition" and its related literature.

The 3rd chapter "Proposed system design" contains information of project design and work flow diagram.

The 4th chapter “System analysis” contains the system limitations, advantages & disadvantages and features & functionalities.

The 5th chapter “Implementation and testing” contains the information about the training of the system and testing.

The 6th chapter “Conclusion” contains a short conclusion of the system and future development of the system.

1.6 Summary

This chapter is about a brief introduction of the project “Design and development of Automatic Bengali Number Plate Recognition (ABNPR) by Supervised Machine Learning” which is important to help improve the traffic management and security. This chapter also includes the motivations, scope and organization of the system.

Chapter 2

Literature Review

2.1 Machine Learning

It is the process of organizing, analyzing and modeling data. Regression and classification is used for predictive modeling. In simple words training the machine to learn. First of we need some data to train the machine after this there is a need to cross validate the learning and at the end we need to test either the results are near to the requirement or not. If the results are near to the requirements it means that our machine is learning well otherwise we need to train the machine again and cross validate again and this process will repeat till our requirement meet.
For learning machine there are number or techniques and many algorithms that we can use.
Here are two types of machine learning techniques:

1. Supervised machine learning.
2. Unsupervised machine learning [23].

2.2 Supervised Machine Learning

Supervised learning (machine learning) takes a known set of input data and known responses to the data, and seeks to build a predictor model that generates reasonable predictions for the response to new data.

Suppose you want to predict if someone will have a heart attack within a year. You have a set of data on previous people, including age, weight, height, blood pressure, etc. You know if the previous people had heart attacks within a year of their data measurements. So the problem is combining all the existing data into a model that can predict whether a new person will have a heart attack within a year.

Supervised learning splits into two broad categories:

1. Classification for responses that can have just a few known values, such as 'true' or 'false'. Classification algorithms apply to nominal, not ordinal response values.
2. Regression for responses that are a real number, such as miles per gallon for a particular car.

Few Supervised classification learning techniques are:

- Logistic regression.
- Generalized linear model.
- Classification.
- Support vector machine.
- Artificial neural network [23].

2.3 Artificial Neural Networks

Neural networks are composed of simple elements operating in parallel. These elements are inspired by biological nervous systems. As in nature, the connections between elements largely determine the network function. You can train a neural network to perform a particular function by adjusting the values of the connections (weights) between elements.

Typically, neural networks are adjusted, or trained, so that a particular input leads to a specific target output. Typically, many input/target pairs are needed to train a network.

Neural networks have been trained to perform complex functions in various fields, including pattern recognition, identification, classification, speech, vision, and control systems.

Artificial neural network is same as human nervous system. In a neural there are three types of layers:

1. Input layer
2. Hidden layers
3. Output layer

Input layer contains inputs data that we need to feed the neural network. Number of inputs depend on the data samples.

Hidden layers contains some hidden nodes and bias nodes. Number of nodes depends on user and complexity of network and can be increased accordingly. Not just nodes but hidden layers can also be increased as per requirements. This network will call multilayer neural network.

Output layer contain output nodes. Number nodes depend on the classes of data number of classes and output nodes will remain same [23].

2.4 Neural Networks Design Steps

We will follow the standard steps for designing neural networks to solve problems in four application areas: function fitting, pattern recognition, clustering, and time series analysis. The work flow for any of these problems has seven primary steps.

1. Collect data.
2. Create the network.
3. Configure the network.
4. Initialize the weights and biases.
5. Train the network.
6. Validate the network.
7. Use the network.

2.5 Object Detection

Object detection models seek to identify the presence of relevant objects in images and classify those objects into relevant classes. Object detection algorithms typically leverage machine learning or deep learning to produce meaningful results. When humans look at images or video, we can recognize and locate objects of interest within a matter of moments. The goal of object detection is to replicate this intelligence using a computer.

2.5.1 Inception ResNet V2

Inception-ResNet-v2 is a convolutional neural network that is trained on more than a million images from the ImageNet database. The network is 164 layers deep and can classify images into 1000 object categories, such as keyboard, mouse, pencil, and many animals. As a result, the network has learned rich feature representations for a wide range of images. The Inception-ResNet-v2 was used for the classification task. The architecture of the network is shown in Figure 4. Inception-Resnet-v2 is formulated based on a combination of the Inception structure and the Residual connection. In the Inception-Resnet block multiple sized convolutional filters are combined by residual connections. The usage of residual connections not only avoids the degradation problem caused by deep structures but also reduces the training time.

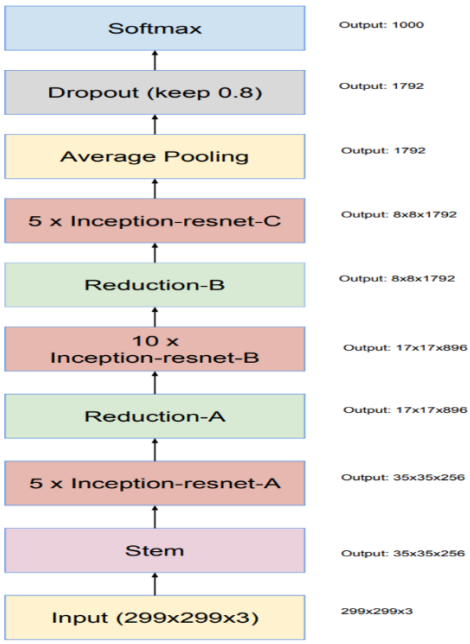


Figure 2.1: Architecture of Inception ResNet V2.

2.5.2 Mean Average Precision

Mean Average Precision (mAP) is a metric commonly used in object detection tasks . It is the average of the average precision (AP) values calculated for each class in the dataset. AP is computed by estimating the area under the precision-recall curve for a single class. The precision-recall curve is obtained by varying the model's confidence score threshold and plotting the precision and recall values . mAP provides an overall measure of the model's performance across multiple classes.

1. mAP formula is based on the following sub metrics:
2. Intersection over Union(IoU),
3. Confusion Matrix,
4. Recall,
5. Precision

1. Intersection Over Union

Intersection over Union (IoU) is a metric that quantifies the degree of overlap between two regions.

As mentioned above, IoU metric evaluates the correctness of a prediction. The value ranges from 0 to 1. With the help of IoU threshold value, we can decide whether a prediction is True Positive**,** False Positive, or False Negative. Interestingly, these terms collectively form the Confusion Matrix**.**

2. Confusion Matrix

To understand the Confusion Matrix, let's take an example of a classification problem. A model has to identify whether there are hot dogs in an image. The predictions can be either correct or incorrect. The following combination of predictions is possible based on the output and the actual image.

Similarly, these terms apply to object detection and segmentation as well. However, the exact meanings are not the same. In object detection, the correctness of the prediction (TP, FP, or FN) is decided with the help of the IoU threshold. Whereas in image segmentation, it is decided by referencing the Ground Truth pixels. Ground truth means known objects.

Note that True Negative is not applicable to object detection and segmentation. It is correctly detecting the background of an object as background. Which is equivalent to not detecting anything.

3. Precision

Precision measures the proportion of predicted positives that are actually correct. If you are wondering how to calculate precision, it is simply the True Positives out of total detections. Mathematically, it's defined as follows.

P = TP/(TP + FP)

= TP / Total Predictions

The value ranges from 0 to 1.

4. Recall

Recall measures the proportion of actual positives that were predicted correctly. It is the True Positives out of all Ground Truths. Mathematically, it is defined as follows.

R = TP / (TP + FN)

= TP / Total Ground Truths

Similar to Precision, the value of Recall also ranges from 0 to 1.

2.6 Optical character recognition (OCR)

Optical character recognition (OCR) software that is used to extract text from the image. Tesseract OCR have a python API and it is open source. Firstly, we will do installation of it. It pretty simple and depend on you OS. Esseract works best when there is a clean segmentation of the foreground text from the background. In practice, it can be extremely challenging to guarantee these types of setups. There are a variety of reasons you might not get good quality output from Tesseract like if the image has noise on the background. The better the image quality (size, contrast, lightning) the better the recognition result. It requires a bit of preprocessing to improve the OCR results, images need to be scaled appropriately, have as much image contrast as possible, and the text must be horizontally aligned.

Chapter 3

Proposed system Design

3.1 Project Architecture

In the below figure ,there are six modules. Labeling, Training, Save Model, OCR and Pipeline, and RESTful API. The process is as follows. First, we will collect the image. Then we have to label images for object detection of License Plate or Number Plate using Image Annotation Tool which is open-source software developed in python GUI. Then after labeling the image we will work on data preprocessing, build and train a deep learning object detection model (Inception Resnet V2) in TensorFlow 2. Once we have done with the Object Detection model training process, then using this model we will crop the image which contains the license plate which is also called the region of interest (ROI), and pass the ROI to Optical Character Recognition API Tesseract in Python (PyTesseract). As well extract text from images. Now, we will put it all together and build a Pipeline Deep Learning model. In the final module, we will create a web app project using FLASK Python. With that, we will be ready with our App.

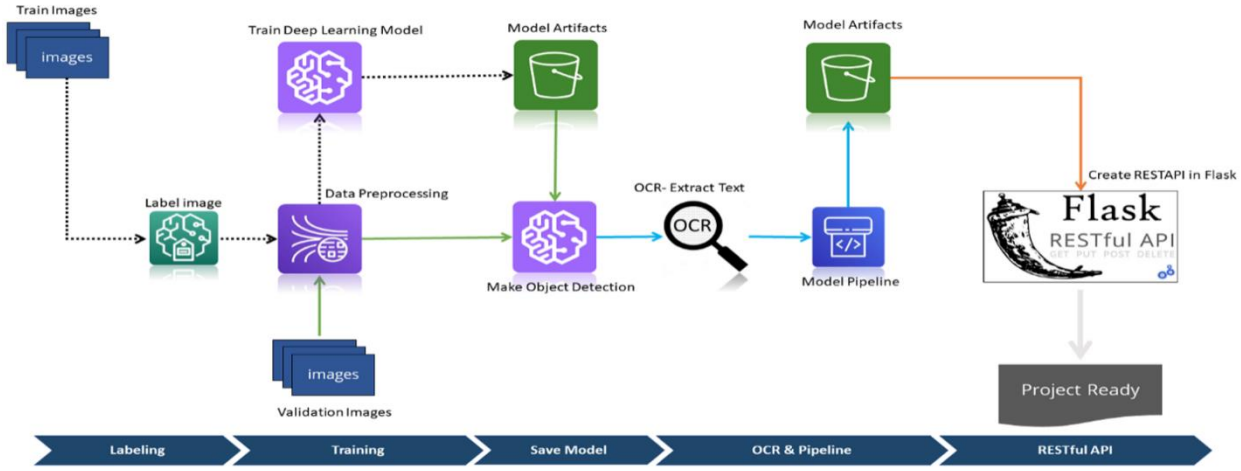


Figure 3.1: Number plate recognition project architecture.

3.1.1 Labeling

For building the number plate recognition we need data. For that, we need to collect the vehicle images where the number plate appears on it as much as we can. For label images, I used

the LabelImg image annotation tool. After that open, the GUI as give the instruction and click on CreateRectBox and draw the rectangle box as shown below and save the output in XML. This is a manual process and you need to do it for all the images. Be careful while doing labeling because the labeling process has a direct impact on the accuracy of the model.

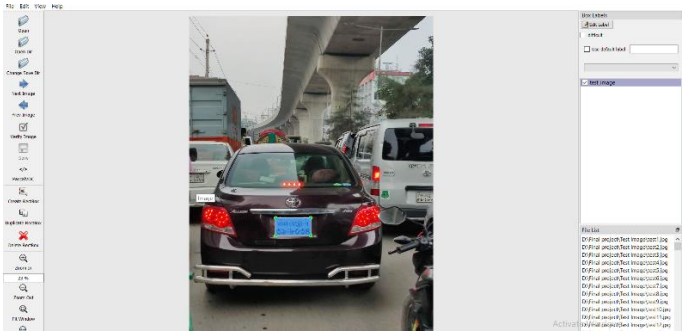


Figure 3.2: Labelimg.

3.1.2 Data processing

Once we are done with the labeling process now we need to do some data preprocessing. Since the output of the label is XML in order to use this for the training process we need data in array format. For that, we will take the useful information from the label which are the diagonal points of the rectangle box or bounding box which are *xmin, ymin, xmax, ymax* respectively. This is available in XML. So we need to extract the information and save it in any convenient format, here I will convert bounding information into CSV and later on, I will convert that into an array using Pandas. Now let's see how to parse the information using Python.

Firstly let me load all Bangladeshi number plate images which I will us in this project at one time. As well I will using xml.etree python library to parse the data from XML and also import pandas and glob. Using glob let first get all the XML files that are produced during labeling.

This is a very important step, in this process we will take each and every image and convert it into an array using OpenCV and resize the image into 224 x 224 which is the standard compatible size of the pre-trained transfer learning model.

After that, we will normalize the image just by dividing with maximum number as we know that the maximum number for an 8-bit image is 2^8 -1 = 255. That the reason we will divide our image 255.0. The way of diving an array with the maximum value is called Normalization (Min-Max Scaler). We also need to normalize our labels too. Because for the deep learning model the output range should be between 0 to 1. For normalizing labels, we need to divide the diagonal points with the width and height of the image. And finally values in a python list. In the next step, we will convert the list into an array using numpy(shown in the figure 12). Now split the data into training and testing set using sklearn. So it will split the data into 90%(for training) and 10%(for testing) as shown in figure 13-

3.1.3 Deep learning for object detection

We are ready to train a deep learning model for object detection. Here we will use the Inception-ResNet-v2 model with pre-trained weights and train this to our data. We are import necessary libraries from TensorFlow-

```
from tensorflow.keras.applications import InceptionResNetV2

from tensorflow.keras.layers import Dense,Dropout,Flatten,Input

from tensorflow.keras.models import Model

import tensorflow as tf
```

Now compile the model and have a look at the summary (Figure 14). The summary is textual and includes information about: The layers and their order in the model, The output shape of each layer, The number of parameters (weights) in each layer.

3.1.4 Inception ResNetV2 supervised model training

To train the Inception-ResNet-V2 model, we need to iterate over the training dataset in multiple epochs and adjust the model's weights based on the loss and optimizer.

By iterating over the dataset in multiple epochs and adjusting the model's weights, the Inception-ResNet-V2 model gradually learns to make better predictions and improve its performance on the training dataset. After that we will save our trained model on our working directory. Then we will look at on scalars on TensorBoard.

3.1.5 Pipeline object detection model

This is the step in object detection where we will use our trained model. In this step, we will put it all together and get the prediction for a given image. First, we would like to try with one of my test pictures of car. Let load our model. Next is loading our TEST picture with right path to it. I loaded some more images for this purpose only – folder Test images. But in order to pass this image of a model, we need to provide the data in the dynamic fourth dimension. We will got the output from the model and output what we got is the normalized output. So, what we need to do is to convert back into our original form values, which actually we did in during the training process, in the training process, we have the original form values and convert that normalized one. So basically, we will de-normalize the values back. Then we will draw bounding box on top of the image. We will provide the two diagonal points. Let’s make use of these points and let’s draw the rectangle box. Lastly what we’ll do, let’s put it all together in one place and create

function. And in the end visualize output. Our output will return image and coordinates of bounding box.

3.1.6 Extract number plate text from image

Firstly, we will load our image and convert to array. Crop our bounding box with coordinates of it. We will identify region of interest (ROI). With use of tesseract, we will extract the text from the image or ROI.

3.1.7 Number plate web app

We will create a web-based user interface by using Flask-Api, Django, Bootstrap, html and CSS. It will use http method for uploading image file as input into our model. We will simply add our pipeline and OCR model, basically all necessary library and function with correct path to model to predict the bounding box, crop it at same time, save the output and show the extracted bengali number which will be shown in our web app output.

3.2 Work Flow Diagram

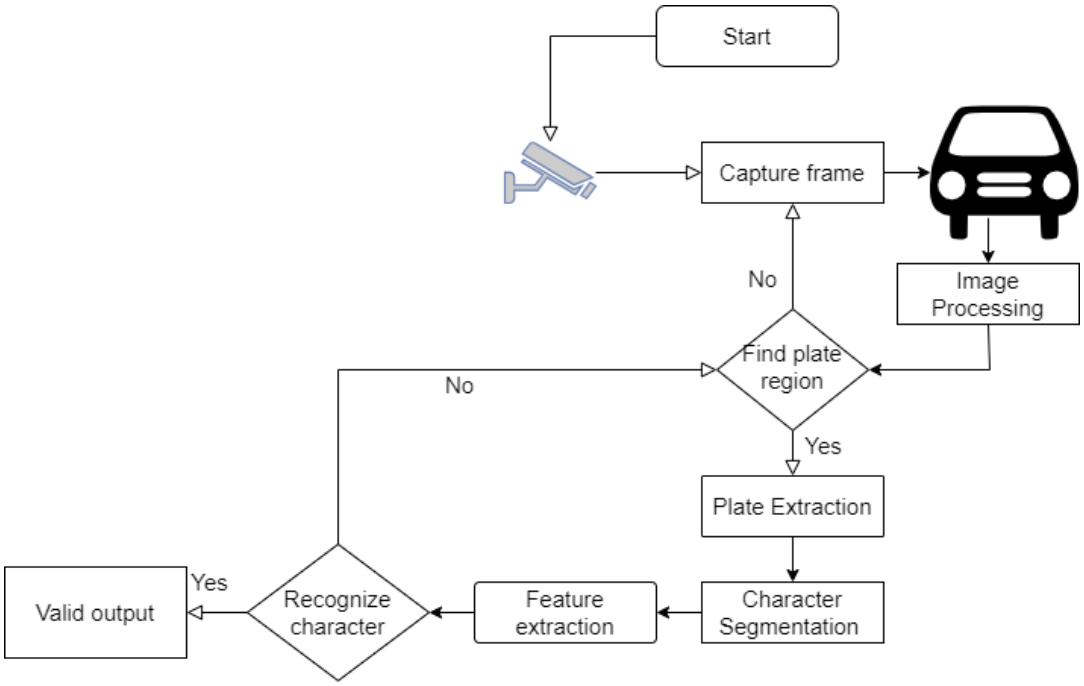


Figure 3.3: Work flow diagram.

Chapter 4

System Analysis

4.1 Existing System limitations

Automatic Number Plate Recognition (ANPR) for Bangladeshi vehicle is a system for automatically identifying the license plates of any vehicle in Bengali script. Some of the limitations of ANPR are:

- It may have difficulty in detecting and reading dirty, damaged, or tilted plates.
- It may be affected by environmental factors such as lighting, weather, or occlusion.
- It may require high-quality cameras and computational resources to achieve high accuracy and speed.

4.2 Features and functionalities

Some of the features and functionalities of Automatic Number Plate Recognition (ANPR) for Bangladeshi vehicle are:

- It can detect and read the license plates of any vehicle in Bengali script.
- It can pre-process the images to enhance the contrast, brightness, and sharpness of the plates.
- It can segment the characters from the plates using edge detection, connected component analysis, or machine learning techniques.
- It can recognize the characters using optical character recognition (OCR), template matching, or deep learning models.
- It can use the recognized plates for various applications such as tracking, ticketing, billing, security, or parking management.

4.3 Advantages and Disadvantages

Some of the advantages and disadvantages of Automatic Number Plate Recognition (ANPR) for Bangladeshi vehicle are:

Advantages:

- It can automate the process of identifying the license plates of any vehicle in Bengali script.
- It can improve the efficiency and accuracy of various applications such as tracking, ticketing, billing, security, or parking management.
- It can reduce the human errors, biases, and costs associated with manual recording and verification of plates.

Disadvantages:

- It may have difficulty in detecting and reading dirty, damaged, or tilted plates.
- It may be affected by environmental factors such as lighting, weather, or occlusion.
- It may require high-quality cameras and computational resources to achieve high accuracy and speed.
- It may raise privacy and ethical concerns regarding the collection and use of personal data from the plates.

4.4 Why choose ABNPR

We may choose Automatic Bengali Number Plate Recognition (ABNPR) for the following reasons:

- It can automatically identify the license plates of any vehicle in Bengali script.
- It can track, ticket, and bill the vehicles based on their license plates.
- It can avoid misuse of the vehicles by verifying their license plates.
- It can identify the Bengali characters in the license plates efficiently and with less computational complexity.

Chapter 5

Implementation and Testing

5.1 Dataset Images

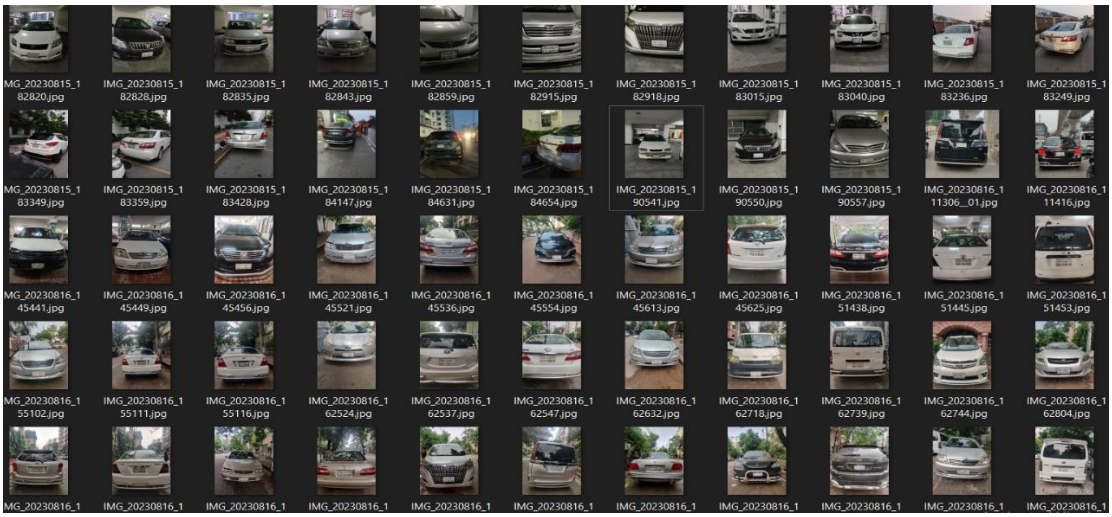


Figure 5.1: Train Images.

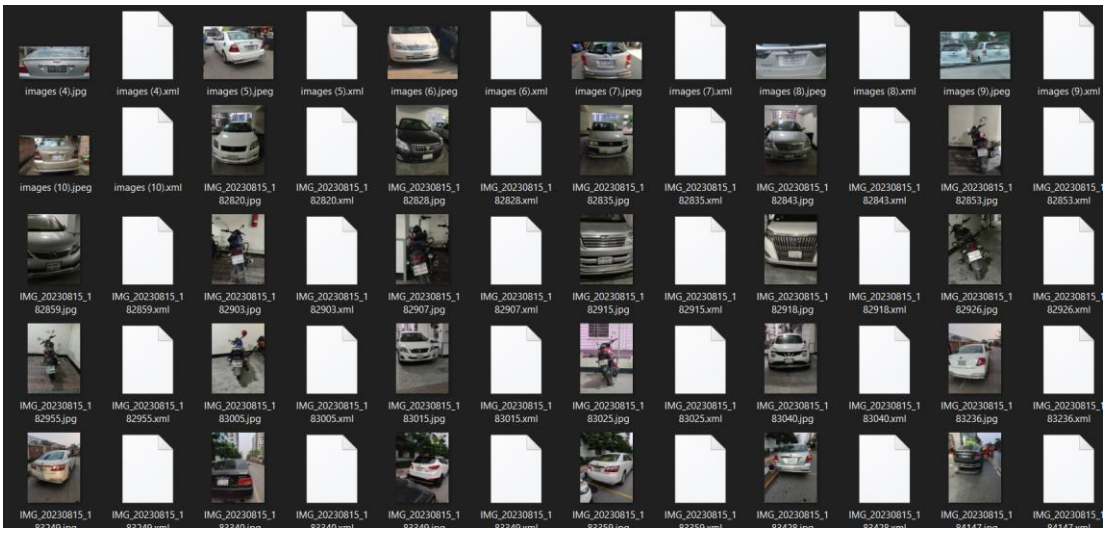


Figure 5.2: Annoted Images.

5.2 Training

We have used Kaggle notebook which provide single 16 GB NVDIA Tesla P100 GPU RAM, for training and evaluating our models. However, we have labeled the image of our dataset for our model in a local machine. Also we use Anaconda prompt and Jupyter notebook in our local machine which have the CPU AMD Ryzen 7 4800H with windows 10 to predict the ROI and extract text from it.

PARSING DATA FROM XML AND CONVERTING IT INTO CSV

We are creating a csv file named as labels.csv where we are saving our every annoted image file path, top left and bottom right's x and y axis coordinates (Figure). Because our Inception ResNet V2 model takes input as .csv file.

	filepath	xmin	xmax	ymin	ymax
0	/kaggle/input/bangladeshi-vehicle-number-plate...	933	1090	2429	2505
1	/kaggle/input/bangladeshi-vehicle-number-plate...	910	1120	2007	2124
2	/kaggle/input/bangladeshi-vehicle-number-plate...	562	884	1976	2341
3	/kaggle/input/bangladeshi-vehicle-number-plate...	1198	1509	2905	3131
4	/kaggle/input/bangladeshi-vehicle-number-plate...	312	1403	2482	3482

Figure 5.3: Data parsing from XML to CSV.

READ THE IMAGE FILE

```
['/kaggle/input/bangladeshi-vehicle-number-plates/number_Plate_raw/IMG_20230725_072211.jpg',
 '/kaggle/input/bangladeshi-vehicle-number-plates/number_Plate_raw/IMG_20230610_175928.jpg',
 '/kaggle/input/bangladeshi-vehicle-number-plates/number_Plate_raw/IMG_20230628_173753.jpg',
 '/kaggle/input/bangladeshi-vehicle-number-plates/number_Plate_raw/IMG_20230713_122106.jpg',
 '/kaggle/input/bangladeshi-vehicle-number-plates/number_Plate_raw/IMG_20230628_173319.jpg',
 '/kaggle/input/bangladeshi-vehicle-number-plates/number_Plate_raw/IMG-20230624-WA0046.jpg',
 '/kaggle/input/bangladeshi-vehicle-number-plates/number_Plate_raw/IMG_20230611_102502.jpg',
 '/kaggle/input/bangladeshi-vehicle-number-plates/number_Plate_raw/IMG_20230628_173220.jpg',
 '/kaggle/input/bangladeshi-vehicle-number-plates/number_Plate_raw/IMG-20230624-WA0008.jpg',
 '/kaggle/input/bangladeshi-vehicle-number-plates/number_Plate_raw/IMG_20230621_115449.jpg']
```

Figure 5.4: Reading image file.

DATA PROCESSING

```
[11…  [array([[[0.49411765, 0.5176471 , 0.5019608 ],
               [0.45882353, 0.4862745 , 0.45882353],
               [0.49411765, 0.5254902 , 0.48235294],
               ...,
               [0.45882353, 0.22745098, 0.10196079],
               [0.42352942, 0.18431373, 0.10980392],
               [0.43137255, 0.22745098, 0.13333334]],

              [[0.49803922, 0.5254902 , 0.49803922],
               [0.49803922, 0.5176471 , 0.49019608],
               [0.49019608, 0.50980395, 0.4862745 ],
               ...,
```

Figure 5.5: Converting python list into an array.

```
# Split the data into training and testing set using sklearn.
x_train,x_test,y_train,y_test = train_test_split(X,y,train_size=0.9,random_state=0)
x_train.shape,x_test.shape,y_train.shape,y_test.shape
```

```
((972, 224, 224, 3), (109, 224, 224, 3), (972, 4), (109, 4))
```

Figure 5.6: Split the data.

DEEP LEARNING FOR OBJECT DETECTION

```
Model: "model"
_________________________________________________________________________________________________
 Layer (type)                Output Shape              Param #   Connected to
=================================================================================================
 input_1 (InputLayer)        [(None, 224, 224, 3       0         []
                             )]

 conv2d (Conv2D)             (None, 111, 111, 32       864       ['input_1[0][0]']
                             )

 batch_normalization (BatchNorm  (None, 111, 111, 32   96        ['conv2d[0][0]']
 alization)                  )

 flatten (Flatten)           (None, 38400)             0         ['conv_7b_ac[0][0]']

 dense (Dense)               (None, 500)               19200500  ['flatten[0][0]']

 dense_1 (Dense)             (None, 250)               125250    ['dense[0][0]']

 dense_2 (Dense)             (None, 4)                 1004      ['dense_1[0][0]']

=================================================================================================
Total params: 73,663,490
Trainable params: 73,602,946
Non-trainable params: 60,544
```

Figure 5.7: Textual summary of compiled model.

Inception ResNet V2 supervised model training

```
Epoch 290/300
98/98 [==============================] - 16s 164ms/step - loss: 1.2020e-04 - val_loss: 0.0080
Epoch 291/300
98/98 [==============================] - 16s 166ms/step - loss: 1.4986e-04 - val_loss: 0.0078
Epoch 292/300
98/98 [==============================] - 16s 165ms/step - loss: 1.0673e-04 - val_loss: 0.0074
Epoch 293/300
98/98 [==============================] - 16s 166ms/step - loss: 9.0398e-05 - val_loss: 0.0076
Epoch 294/300
98/98 [==============================] - 16s 164ms/step - loss: 5.7724e-05 - val_loss: 0.0075
Epoch 295/300
98/98 [==============================] - 16s 166ms/step - loss: 6.9727e-05 - val_loss: 0.0076
Epoch 296/300
98/98 [==============================] - 16s 164ms/step - loss: 6.2240e-05 - val_loss: 0.0067
Epoch 297/300
98/98 [==============================] - 16s 167ms/step - loss: 5.2566e-05 - val_loss: 0.0069
Epoch 298/300
98/98 [==============================] - 16s 164ms/step - loss: 3.4862e-05 - val_loss: 0.0067
Epoch 299/300
98/98 [==============================] - 16s 166ms/step - loss: 2.6592e-05 - val_loss: 0.0072
Epoch 300/300
98/98 [==============================] - 16s 164ms/step - loss: 3.1105e-05 - val_loss: 0.0069
```

Figure 5.8: Model training.

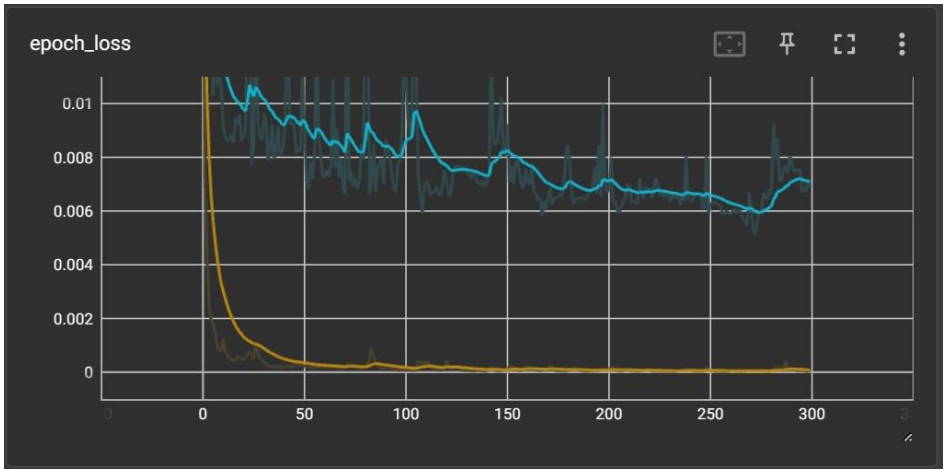


Figure 5.9: Epoch loss graph.

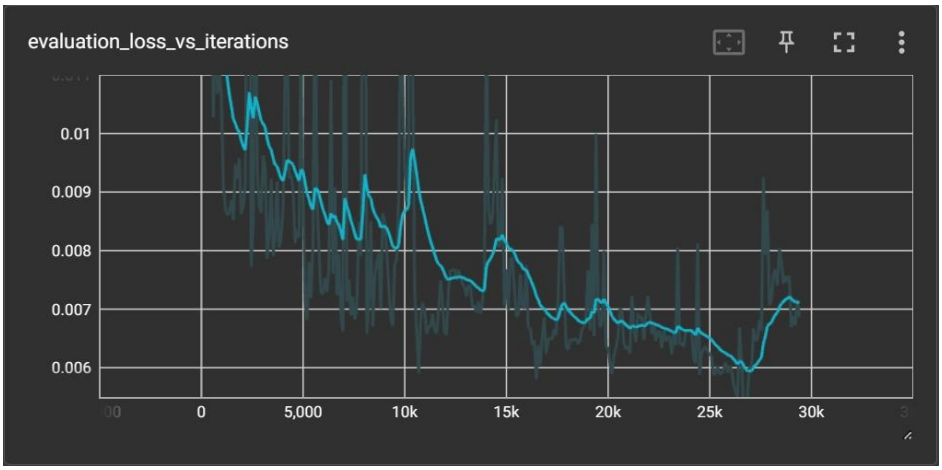


Figure 5.10: Evaluation loss vs iterations graph.

Pipeline object detection model

Make Prediction



Figure 5.11: Making prediction.

5.3 Results

Table 1

Performances of different models on localizing number plate from images:

	Model							
Metric	EfficientDet D0	EfficientDet D1	EfficientDet D2	SSD MobileNet 320 Lite	SSD MobileNet 640	SSD ResNet 640	YOLO v4	Inception ResNet v2
mAP	0.9736	0.9956	0.9831	0.9871	0.9451	0.9843	0.9937	0.9731
Recall	0.7393	0.7248	0.6854	0.7665	0.7111	0.7579	0.9800	0.9600

Table 2

Performances of different models on text recognition from images:

	Model							
Metric	EfficientDet D0	EfficientDet D1	EfficientDet D2	SSD MobileNet 320 Lite	SSD MobileNet 640	SSD ResNet 640	YOLO v4	Inception ResNet v2
mAP	0.9089	0.9184	0.8940	0.9254	0.9448	0.9359	0.9631	0.9356
Recall	0.6869	0.7334	0.7127	0.7497	0.7477	0.7437	0.9800	0.9600

5.4 GUI for proposed system

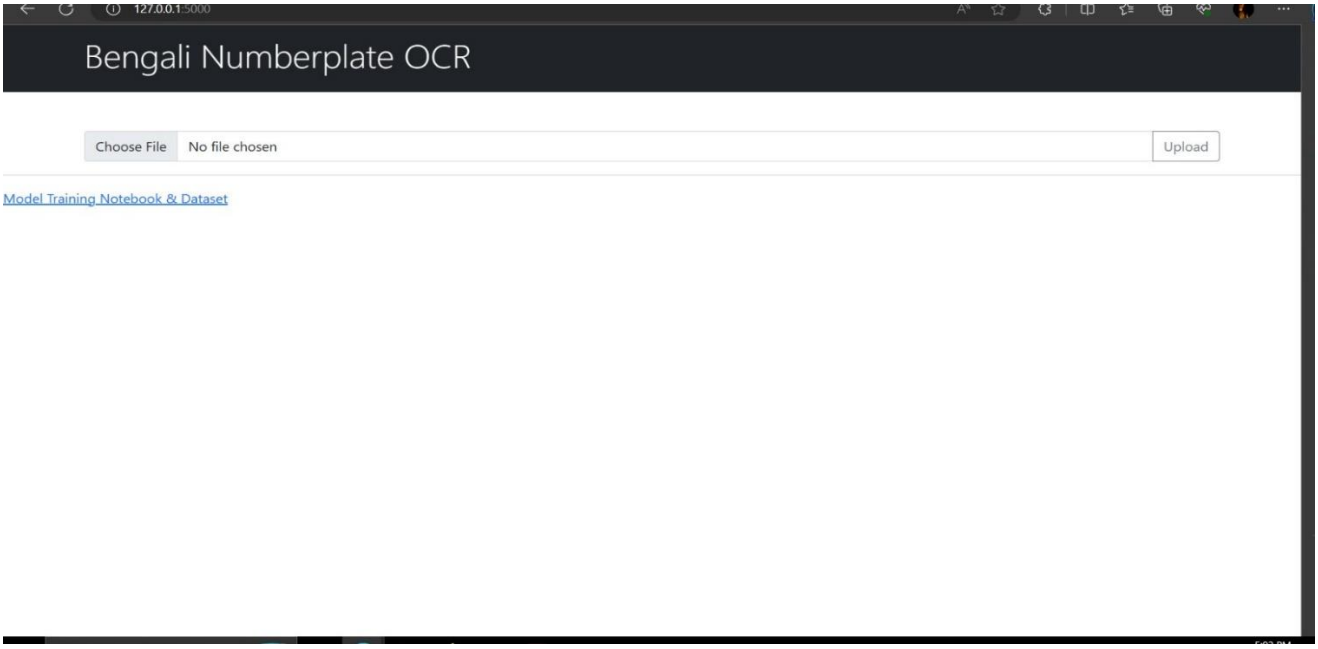


Figure 5.12: User interface.

5.5 System testing

User interface testing of ABNPR

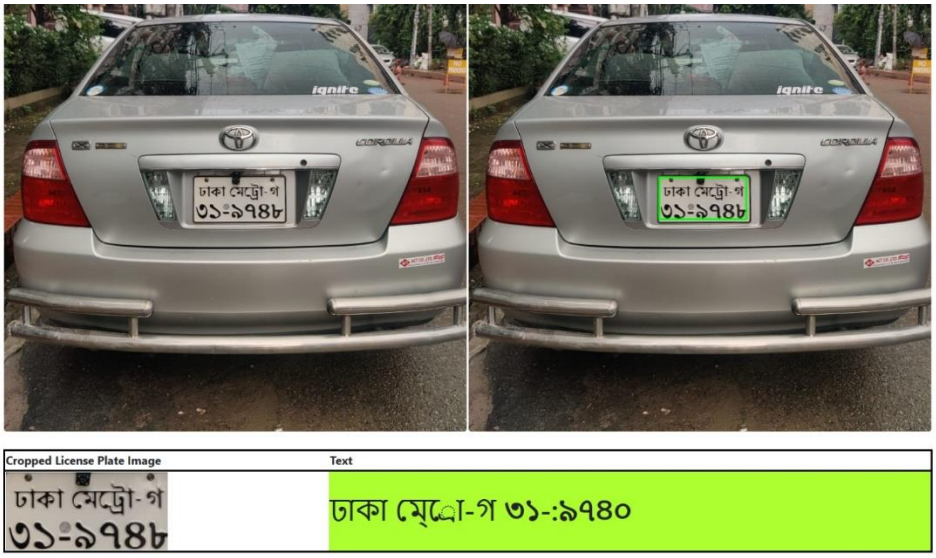


Figure 5.13: OCR in user interface.

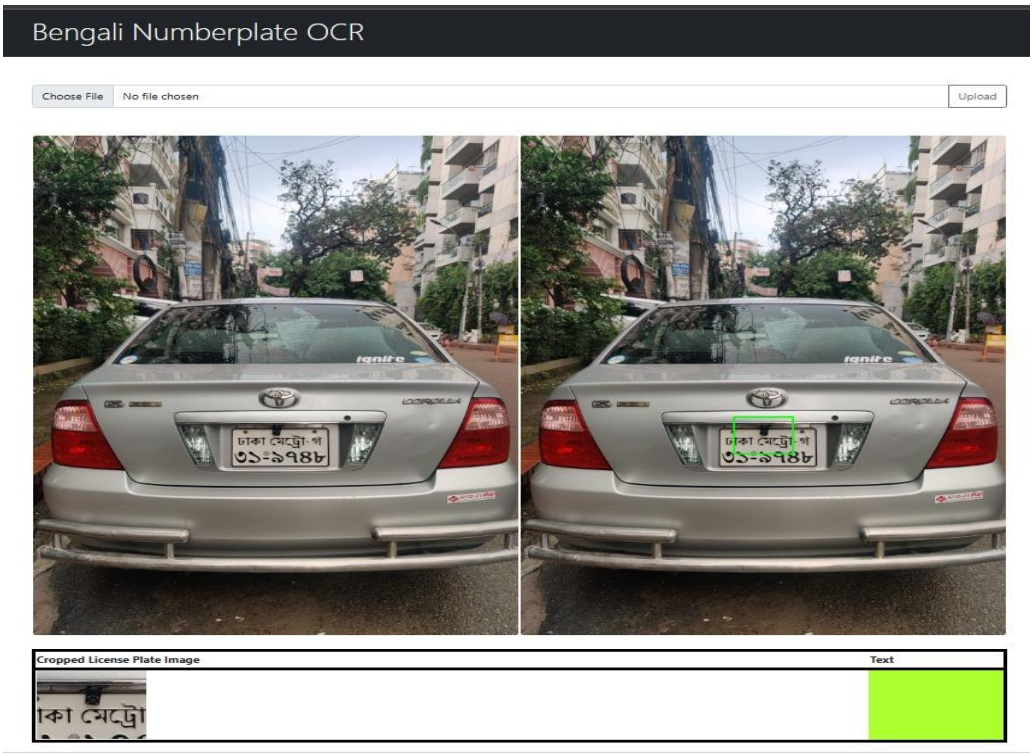


Figure 5.14: OCR in user interface.

Chapter 6

Conclusion

6.1 Scope for future development

- **Improving the accuracy and robustness of the system:** Developing more advanced image processing and machine learning techniques to handle various challenges such as low-quality images, different lighting conditions, occlusions, distortions, etc.
- **Expanding the scope of the system:** Applying the system to other types of vehicles such as motorcycles, buses, trucks, etc. and other languages such as Arabic, Hindi, Urdu, etc.
- **Integrating the system with other applications:** Connecting the system with other databases and systems such as traffic management, law enforcement, toll collection, parking management, etc.
- **Enhancing the security and privacy of the system:** Developing more secure and encrypted methods to store and transmit the license plate data and prevent unauthorized access or misuse.
- **Reducing the cost and complexity of the system:** Developing more efficient and affordable hardware and software components to implement the system and make it more accessible and scalable.
- **Evaluating the social and ethical impacts of the system:** Assessing the potential benefits and risks of the system for different stakeholders such as vehicle owners, drivers, authorities, etc. and ensuring the system is fair, transparent, and accountable.

6.2 Summary

Automatic Bengali Number Plate Recognition (ABNPR) is a system that can automatically detect and read the license plates of vehicles written in Bengali script. It uses various image processing and machine learning techniques to identify the characters and digits on the plates. ABNPR can be useful for various applications such as tracking, ticketing, and billing.

Capturing the image of the vehicle's license plate using a camera or a video. Enhancing the quality of the image by applying filters, contrast adjustment, noise reduction, etc.

Locating the region of interest (ROI) that contains the license plate in the image using methods such as edge detection, connected component analysis, or deep learning models such as Inception ResNet V2.

Separating the individual characters and digits on the license plate using methods such as thresholding, contour detection, or deep learning models such as CNN.

Recognizing the characters and digits on the license plate using methods such as optical character recognition (OCR), template matching, or deep learning models such as CNN.

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